Lab Report: Model comparison on benchmarks

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 30 epochs with a batch size of 32. Each experiment was repeated 5 times with different random seeds.

2 Results

Figure 1 shows the accuracy during training.

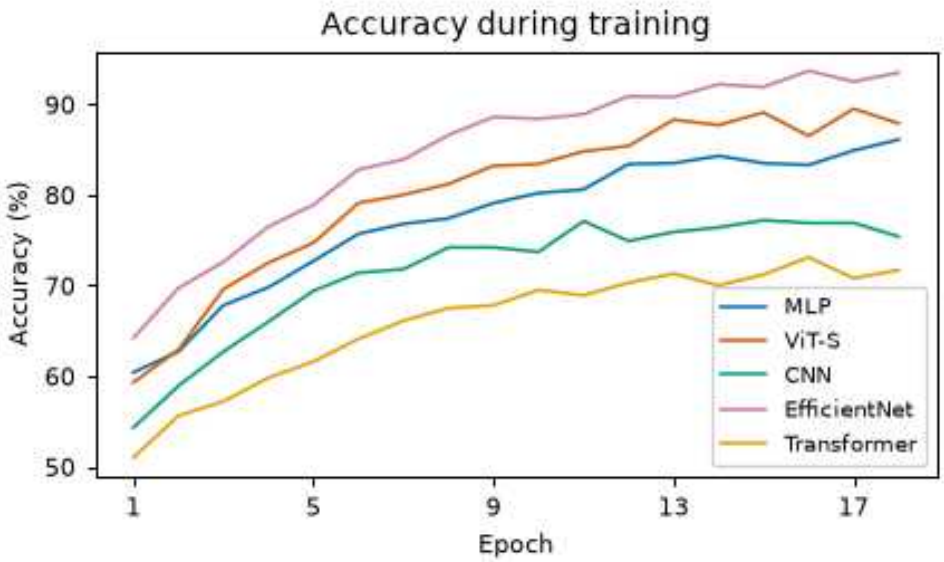


Figure 1: Accuracy during training for the models in this experiment.

As shown in Figure 1, ViT-S attains a accuracy of 57.7% at epoch 8. As shown in Figure 1, EfficientNet attains a accuracy of 92.8% at epoch 15.

Figure 2 shows the f1 score during training.

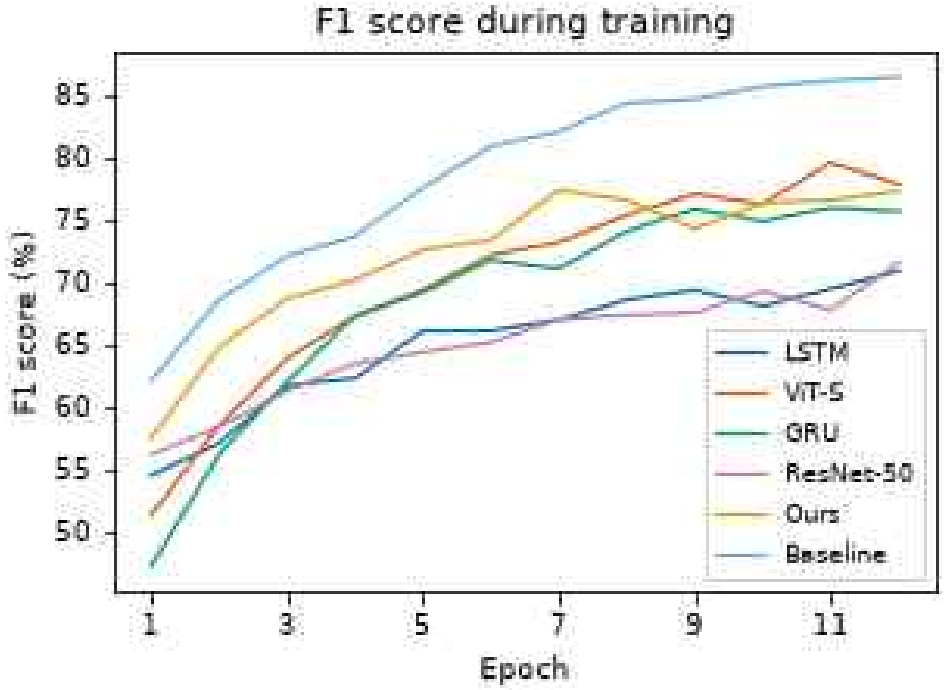


Figure 2: F1 score during training for the models in this experiment.

From epoch 8 to epoch 11, the F1 score of Baseline improves by a relative 3.2%. As shown in Figure 2, Ours

outperforms ResNet-50 by 9.7 percentage points at epoch 11. Ours reaches 74% F1 score at epoch 9 (Figure 2). As shown in Figure 2, at epoch 5, ResNet-50 achieves 65.2% F1 score.

Figure 3 shows the training loss.

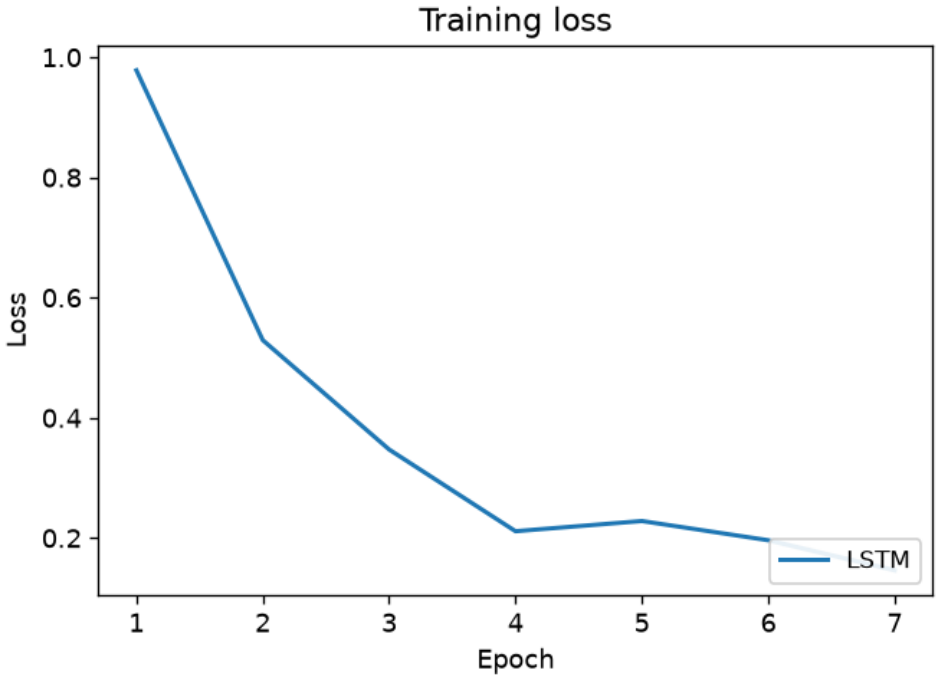


Figure 3: Training loss for the models in this experiment.

LSTM has a loss of 0.288 at epoch 6. At epoch 5, the loss of LSTM is 0.228. LSTM has a loss of 0.621 at epoch 2.

Figure 4 shows the accuracy of Ours by dataset.

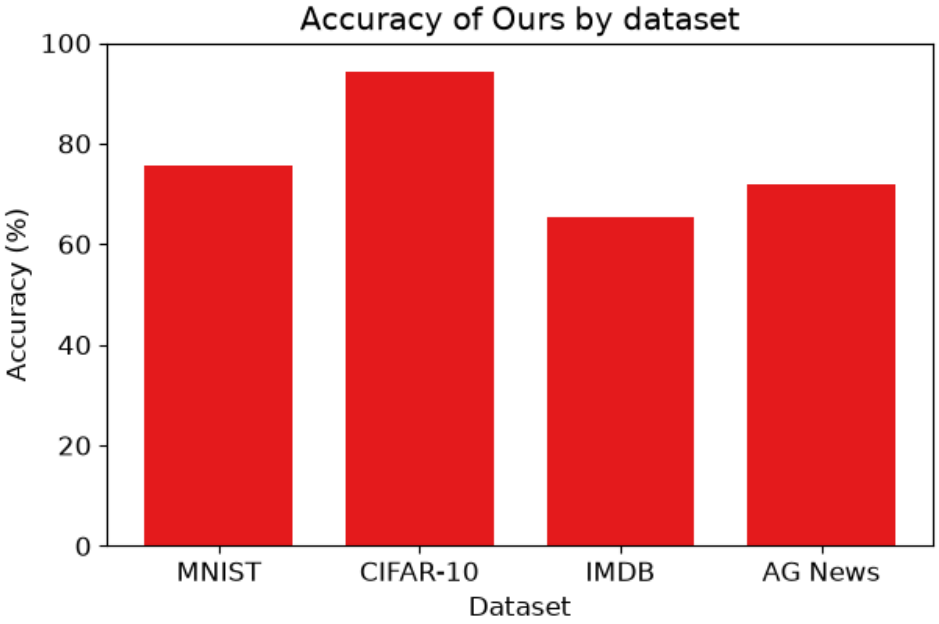


Figure 4: Accuracy of Ours by dataset for the models in this experiment.

Ours scores 13.8 points higher on MNIST than on AG News. Ours scores 29 points higher on CIFAR-10 than on IMDB. On IMDB, Ours obtains a accuracy of 94.4%.

3 Discussion

On STL-10, Naive Bayes obtains an accuracy of 90.3%. Overall, the results are in line with what we expected

from the literature.